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Google Search Indicators and Nowcasting German Unemployment: Early vs. Late Information

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Abstract

This seminar paper evaluates whether Google search data can improve nowcasts of German unemployment. Following a biweekly aggregation framework, weekly search indicators for 2011–2023 are transformed into early-month and previous-month components to reflect the timing of the German unemployment measurement process. After confirming that both unemployment and search series are nonstationary, the analysis is conducted in first differences using rolling-window AR and ARX benchmarks as well as mixed-frequency MIDAS regressions. Pseudo real-time nowcasts over 2019–2023, including the COVID-19 shock, are evaluated using RMSE, MAE and Diebold–Mariano tests. Results show that German unemployment is highly persistent and that Google search indicators provide only limited incremental predictive power beyond simple AR dynamics. Weekly MIDAS models capture slightly more high-frequency variation than biweekly ARX specifications but do not substantially outperform the AR benchmark. Moreover, early-month and previous-month search components are statistically indistinguishable in terms of forecast accuracy. Overall, the informational content of Google search data for German unemployment appears modest and sensitive to the chosen temporal aggregation.

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1 Introduction

Timely and reliable macroeconomic indicators are essential for economic research institutions and policymakers, particularly central banks, as they assess the stance of fiscal and monetary policy (Ferrara and Simoni, 2020; Bańbura et al., 2013). However, in many countries key statistics, most notably unemployment, are published only with considerable delay after the reference period. Such publication lags create informational gaps that can impede effective decision-making, especially during periods of heightened uncertainty or rapid economic change. Episodes such as the COVID-19 pandemic or the 2008 financial crisis demonstrated how delays in labor-market and consumption indicators complicated the ability of authorities to respond swiftly to emerging developments (Chetty et al., 2023). These challenges stem largely from the nature of traditional data collection. Survey-based and administrative statistics are often resource-intensive and therefore cannot capture economic dynamics in real time.

Against this backdrop, researchers and institutions have increasingly turned to high-frequency, unconventional data sources to complement official statistics and improve short-term forecasting. Digital traces such as job-posting data, mobility patterns, and, most prominently, search-engine activity, offer new possibilities for real-time monitoring (Barbaglia et al., 2023). These data sources reflect aspects of individual behavior and expectations at a much higher frequency than traditional statistics.

Among these alternatives, search-engine queries stand out due to their ubiquity and their strong link to information-seeking behavior. Search engines constitute the primary interface through which users access online information. Furthermore, they are used across virtually all demographic and socioeconomic groups for purposes ranging from job search to product research and education. Consequently, search data provide large-scale behavioral signals that reveal what individuals are currently concerned with. Google’s dominant market position ensures that Google Trends offers broad and demographically diverse coverage of search behavior.

The theoretical appeal of search data lies in their behavioral immediacy. Individuals typically conduct online searches before engaging in economic actions, such as applying for jobs or adjusting consumption behavior. For example, rising search volumes for terms like “job center,” “layoff,” or “unemployment benefits” may indicate emerging stress in the labor market before it becomes visible in official statistics. When aggregated across millions of users, such search activities form a micro-level digital footprint that can serve as a high-frequency proxy for macroeconomic conditions. In this sense, search data operate as a near real-time indicator of economic sentiment and intention, making them particularly suited for nowcasting applications where official data are published with a delay.

The idea that digital search behavior captures real-time shifts in economic sentiment has inspired a series of empirical studies seeking to quantify its predictive value. Early

contributions in this area laid the groundwork for using online search data as an economic indicator.

One of the earliest contributions to this line of research comes from Ettredge et al. (2005) who examined whether search volumes from WordTracker’s metasearch engine could anticipate U.S. unemployment. Although their short-term signals were noisy, they showed that smoothed, longer-run trends in job-related searches were strongly associated with future unemployment. This early work established the core intuition that aggregated online search activity captures evolving labor-market expectations before they appear in official data releases.

Building on this foundation, Askitas and Zimmermann (2009) conducted the first systematic attempt to quantify the informational content of Google search queries for German macroeconomic forecasting. Using German data from 2004–2009, they demonstrated that job-related searches reliably lead unemployment releases by several weeks. Their results established Google Trends data as a practical, publicly available high-frequency proxy for labor-market stress.

A major conceptual advance came from Choi and Varian (2012), who extended the use of GT data beyond unemployment to a broad set of macroeconomic indicators. Rather than focusing on multi-step forecasting, they emphasized nowcasting, using contemporaneous search information to fill in publication delays. Their autoregressive models augmented with search volumes consistently improved out-of-sample performance, especially around turning points.

Subsequent research tested the robustness of this approach across different economic environments. For instance, Carrière-Swallow and Labbè (2010) showed that search-based indices can predict automobile sales in Chile, improving both in-sample and out-of-sample accuracy while capturing market reversals earlier than official statistics.

D’Amuri and Marcucci (2010) evaluate whether a Google Job Search Index improves U.S. unemployment forecasts relative to traditional indicators. Comparing a large set of autoregressive benchmark models with search-augmented specifications and conducting an extensive forecast evaluation, they find that models including Google search variables outperform traditional benchmarks in the majority of U.S. states.

At the same time, research broadened beyond labour markets: Goel et al. (2010) use search data from Yahoo! to predict consumer demand for entertainment products, showing that search intensity can strongly correlate with future sales, particularly in rapidly evolving markets.

Beyond large Western economies, Pavlicek and Kristoufek (2014) examine the link between job-related Google searches and unemployment rates for the Visegrad Group countries. Their nowcasting framework nests a simple autoregressive specification in unemployment changes and an augmented model that includes several lags of changes in search intensity. In a pseudo out-of-sample exercise for 2012–2013, the Google-augmented model signif-

icantly outperforms the benchmark for the Czech Republic and Hungary, while results are mixed for Poland and the baseline model even dominates for Slovakia. These findings highlight that the predictive value of search indicators can be substantial even for small and medium-sized economies, but is far from universal.

More recently, studies have increasingly combined high-frequency search data with mixed-frequency econometric techniques, enabling real-time estimation of monthly figures using daily or weekly information. For instance, Costa et al. (2024) propose a MIDAS framework for nowcasting Portuguese unemployment using daily GT series, overcoming historical GT sampling constraints through overlapping-window collection and temporal disaggregation. Their results show that daily GT predictors substantially outperform both monthly GT predictors and benchmark models, especially during economically volatile periods such as COVID-19.

Despite the large international literature on search-based economic forecasting, few recent studies re-examine the German labour market using up-to-date Google Trends data, especially during recessionary episodes such as the COVID-19 shock.

This seminar paper revisits the approach of Askitas and Zimmermann (2009) for Germany and combines it with insights from recent MIDAS applications. From Askitas and Zimmermann (2009) I adopt the biweekly aggregation framework that aligns Google search data with the measurement and publication schedule of unemployment statistics. Specifically, I construct a monthly panel for 2011–2023 that merges official German unemployment with biweekly Google Trends indicators for an early-month window and a previous-month window. From the Portuguese MIDAS literature, I take the idea of exploiting the underlying high-frequency search series directly, using unrestricted MIDAS regressions with weekly Google Trends data.

Within this setting, I address three questions. First, do biweekly Google search indicators improve real-time nowcasts of German unemployment relative to a simple autoregressive benchmark? To answer this, I estimate AR(1) and ARX models in first differences, where the ARX specification augments unemployment persistence with contemporaneous biweekly search intensity. Second, does biweekly aggregation discard useful predictive information contained in the weekly data? I investigate this by comparing the biweekly ARX models with unrestricted MIDAS regressions that incorporate several weekly lags of the underlying search series. Third, following Askitas and Zimmermann (2009), I ask whether early-month or previous-month search activity is more informative for nowcasting. All models are evaluated in a pseudo real-time nowcasting exercise over 2019–2023, using standard forecast accuracy measures and Diebold–Mariano tests. The results provide updated evidence on the limitations of Google search indicators for monitoring short-run labour-market dynamics in Germany.

The remainder of the paper is structured as follows. Section 2 describes the data and construction of the Google Trends indicators. Section 3 presents descriptive statistics

and time-series properties. Section 4 outlines the econometric models and nowcasting framework. Section 5 reports the empirical results and Section 6 concludes.

2 Data

2.1 German Official unemployment series

The target variable of this seminar is the monthly, seasonally unadjusted unemployment rate published by the *Bundesagentur für Arbeit*. Following Askitas and Zimmermann (2009), I use the *Arbeitslosenquote bezogen auf alle zivilen Erwerbspersonen*, which measures unemployment relative to the civilian labour force and is reported in percentage points. Using the unadjusted series is deliberate. Seasonal patterns such as winter layoffs, school-to-work transitions or holiday effects carry information that may be reflected in high-frequency search activity. Removing these variations ex ante would risk discarding precisely the fluctuations Google Trends may capture. Seasonality will later be controlled for explicitly through monthly dummy variables rather than through pre-adjustment.

A key institutional feature is the timing of unemployment measurement and publication. Although figures are released at the end of month, they refer to labour-market conditions up to the middle of that month. This reporting lag creates a natural real-time forecasting environment: at the time a forecaster forms a nowcast for month M , the latest available official observation refers to month $M - 1$. Search data from different parts of the month may therefore contain distinct information sets depending on whether they precede or follow the publication of the previous unemployment number.

The sample used in this study covers May-2011 to January 2023. This provides more than a decade of monthly observations and a sufficiently long evaluation window for the out-of-sample nowcasting exercise. The starting date is chosen for both statistical and economic reasons. On the data side, early Google search data were collected via the now-discontinued Google Insights interface, and several methodological revisions before 2010 make the pre-2011 period difficult to compare consistently with modern Google Trends indices. Internet usage also increased substantially during the 2000s, implying that early search volumes may not yet reflect broad population behaviour.

From an economic perspective, starting the sample in 2011 allows the analysis to focus on a relatively stable period of the German labour market, while still covering several distinct macroeconomic phases. The chosen window includes the post-Eurozone crisis recovery, the pre-pandemic unemployment lows, the COVID-19 shock, and the inflationary period after 2022. These distinct episodes offer a rich environment for evaluating real-time predictive performance.

While excluding the earlier 2004–2009 period prevents an exact replication of the original sample in Askitas and Zimmermann (2009), the improved data quality and economic

comparability of the 2011–2023 period make it more suitable for the nowcasting analysis in this seminar.

2.2 Google trends data

Google Trends provides anonymized indicators of search activity by reporting, for each keyword and region, a 0–100 index of relative search intensity. The index does not reflect absolute search volumes; instead, it is normalized such that 100 corresponds to the peak share of that keyword within the selected time window, and all other values express proportions of that peak. As a consequence, the data measure relative interest and series for different keywords are not comparable in levels unless transformed or standardized.

A second implication is that Google internally relies on sampled query data. Repeating the same download on different days can therefore yield slightly different values (Carrière-Swallow and Labbè, 2010). While some studies mitigate this by averaging multiple repeated downloads, I rely on single downloads due to time constraints.

Because the index is normalised within each individual query and time window, Google Trends series must be preprocessed before they can be used in time-series models that require consistent scaling across long samples, such as the ARX and unrestricted MIDAS regressions employed in this study. In the next subsection, I therefore construct longer, consistently scaled series by stitching overlapping Google Trends queries.

2.3 Constructing of a consistent weekly Google Trends History

Google Trends provides weekly data only for horizons of about five years. To obtain a continuous weekly series for 2011–2023, I will use the following stitching approach: multiple overlapping windows are downloaded, and the later windows are rescaled to match the earlier ones in their overlap. Let $x_t^{(j)}$ denote the weekly Google Trends index for a given keyword in window j at week t , and $x_t^{(j+1)}$ the corresponding index from the adjacent window $j + 1$ over their overlapping period $\mathcal{T}_{j,j+1}$. For any two adjacent windows j and $j + 1$, a scaling factor is computed as the median ratio of their overlapping weekly values,

$$s_{j+1 \leftarrow j} = \text{median} \left\{ \frac{x_t^{(j)}}{x_t^{(j+1)}} : t \in \mathcal{T}_{j,j+1} \right\}, \quad (1)$$

and the later window is multiplied by this factor before concatenation.¹ This procedure reconciles the window-specific 0–100 normalisations and yields a coherent weekly series for each keyword. The resulting data remain relative indices, but they are now internally consistent over time, which is essential for the MIDAS models that use the weekly

¹In practice, I compute the median ratio only over overlapping weeks where both indices are strictly positive, to avoid distortions from zero values.

information directly. A detailed illustration of the stitching procedure for one keyword, including the overlap and rescaling, is provided in Appendix 10.

2.4 Biweekly aggregation approach

To closely follow the biweekly framework of Askitas and Zimmermann (2009), I construct Google Trends indicators that mirror the timing structure of German unemployment measurement. The Federal Employment Agency publishes the unemployment rate for month M at the end of that month, although the reference date lies around the middle of month M . Search behaviour may therefore reflect earlier information about economic conditions in month M .

The key motivation for the biweekly split originates from Askitas and Zimmermann (2009). The unemployment rate for month $M-1$ is released at the *end of month* $M-1$, which is approximately between the two biweekly intervals. As a result, search activity in the first half of month M ($W12_M$) may react to the publication of U_{M-1} , for example due to media coverage or individual responses to newly announced labour-market figures. By contrast, search activity in the second half of month $M-1$ ($W34_{M-1}$) is affected only by the earlier release of U_{M-2} and cannot respond to U_{M-1} . In this sense, $W34_{M-1}$ is expected to contain “cleaner” and more forward-looking information about the labour market than $W12_M$, which may be contaminated by news effects (Askitas and Zimmermann, 2009). To approximate these two information sets for a given reference month M , I build two half-month aggregates from the weekly series:

- $W12_M$: the average of weekly search indices whose Sunday timestamp falls in the first half of month M (days 1–15). This window occurs *after* the publication of unemployment for $M-1$ and overlaps with the measurement date for month M .
- $W34_{M-1}$: the average of weekly search indices whose Sunday timestamp falls in the second half of month $M-1$ (days 16–end of month), aligned forward to month M . This window captures *earlier* search signals that precede the measurement of unemployment in month M and are not influenced by the release of U_{M-1} .

Unlike Askitas and Zimmermann (2009), who proportionally split weekly observations at intra-week boundaries to construct half-month aggregates, I assign each weekly observation to the half-month containing its Sunday timestamp. This avoids mechanically splitting individual weekly observations across months and yields a simple, reproducible mapping from the weekly series to biweekly indicators. The resulting biweekly panel spans June 2011 to January 2023, yielding $T = 140$ monthly observations. This horizon provides sufficient degrees of freedom for time-series estimation while allowing a clear separation between *earlier* and *later* search signals relative to each reference month.

2.5 The selection of google trends variables

The predictive value of search data depends critically on the choice of keywords (Choi and Varian, 2012). Google Trends reflects behavioural intentions rather than economic quantities, and unsuitable keywords can introduce news-driven noise or capture unrelated dynamics. It is therefore important to select search terms that plausibly proxy job-search and unemployment-related behaviour.

In the original study of Askitas and Zimmermann (2009), search behaviour was grouped into four conceptual categories: (k1) contacts with the unemployment office, (k2) interest in the unemployment rate itself, (k3) activity around personnel consulting, and (k4) searches related to job boards. In this paper, I focus on a smaller, coherent set of modern search terms that align with the original k1, k3 and k4 concepts while deliberately excluding k2-type keywords, such as “Arbeitslosenquote”, that primarily capture attention to published statistics rather than job-search behaviour.

The term *arbeitsamt* remains a colloquial and widely used expression associated with unemployment services. It likely captures information seeking related to registration, benefit claims and interactions with the public employment service, and thus serves as a proxy for impending or actual inflows into unemployment. This term corresponds closely to the original k1 category.

The keyword *Jobbörse* refers to the official online job portal of the Federal Employment Agency. Search activity for this term reflects use of a central institutional job-search channel that is tightly connected to unemployment services. It is part of the k4 job-board category as it represents a conceptually clean public-platform signal over the 2011–2023 period.

The terms *stepstone* and *indeed* represent a share of the dominant private online job-search platforms in Germany after 2010. Including them captures a substantial share of modern job-search behaviour. These keywords update the original k4 category to a more current set of platforms while avoiding earlier job-board brands that have since lost relevance.

Finally, the keyword *bewerbung* (job application) encompasses searches related to writing cover letters, preparing CVs and the application process more generally. It reflects a broader job-search intention at the employment transition margin and is behaviourally related to the k3 category in Askitas and Zimmermann (2009). Together, this set of five keywords is intended to capture both interactions with public employment services and modern online job-search activity without relying on purely news-driven terms.

2.6 Restrictions

Before turning to the modelling, it is important to acknowledge several limitations of Google Trends data that affect interpretation and forecasting accuracy.

First, Google Trends is based on sampled query data rather than the full universe of

searches, so identical requests on different days can yield slightly different series (Carrière-Swallow and Labbè, 2010). In addition, Google applies an internal thresholding rule for low-volume queries; although the exact cutoff is unknown, it may suppress or smooth observations in periods of weak search intensity. Both features introduce measurement noise, which is particularly relevant for high-frequency models such as unrestricted MIDAS regressions, and imply that Trends data should be interpreted as noisy behavioural proxies rather than precise measurements.

Second, search terms may carry multiple meanings or evolve over time. Although the selected keywords in this study are behaviourally well defined, they may still capture heterogeneous intentions (for example, *bewerbung* searches by students, career changers, or unemployed individuals). Google’s query classification system is not observable to the researcher, and keyword-based indices cannot distinguish between different user groups or contexts in which the terms are used.

Third, Google Trends does not provide demographic or labour-force identifiers due to privacy constraints. As emphasised by D’Amuri and Marcucci (2010), search activity reflects the behaviour of both unemployed and employed individuals. Because unemployed job search tends to be countercyclical, while on-the-job search is typically procyclical, the resulting aggregate index may blur the distinction between labour-market states.

Finally, weekly data do not align perfectly with half-month intervals, so some months contain uneven numbers of observations in the W12 and W34 segments. Assigning each weekly observation to the half-month containing its Sunday timestamp introduces small timing mismatches in the biweekly indicators.

Overall, these limitations imply that Google Trends indicators should be interpreted as timely but noisy behavioural signals.

3 Descriptive statistics and visual inspection

3.1 Descriptives

In this section, I take a structured look at the biweekly search indicators in order to (i) verify that the aggregation behaves coherently over time and (ii) document broad comovement patterns between the search indicators and unemployment. Figure 1 plots both components for each keyword. Basic descriptive statistics for the biweekly indicators and unemployment are reported in Table 1. The levels of the biweekly indices are broadly comparable across W12 and W34_{prev} for each keyword, with means typically between 40 and 65 index points. The private portal *indeed* exhibits the highest variability, reflecting their strong growth and pronounced cyclicity over the sample period, whereas the unemployment rate fluctuates in a relatively narrow band around an average of 6%.

Across all series, the most notable feature is the strong and highly regular seasonality.

Public-sector related terms such as *arbeitsamt*, *jobbörse* and *bewerbung* exhibit pronounced intra-year cycles with peaks around the turn of the year and troughs in the summer months.

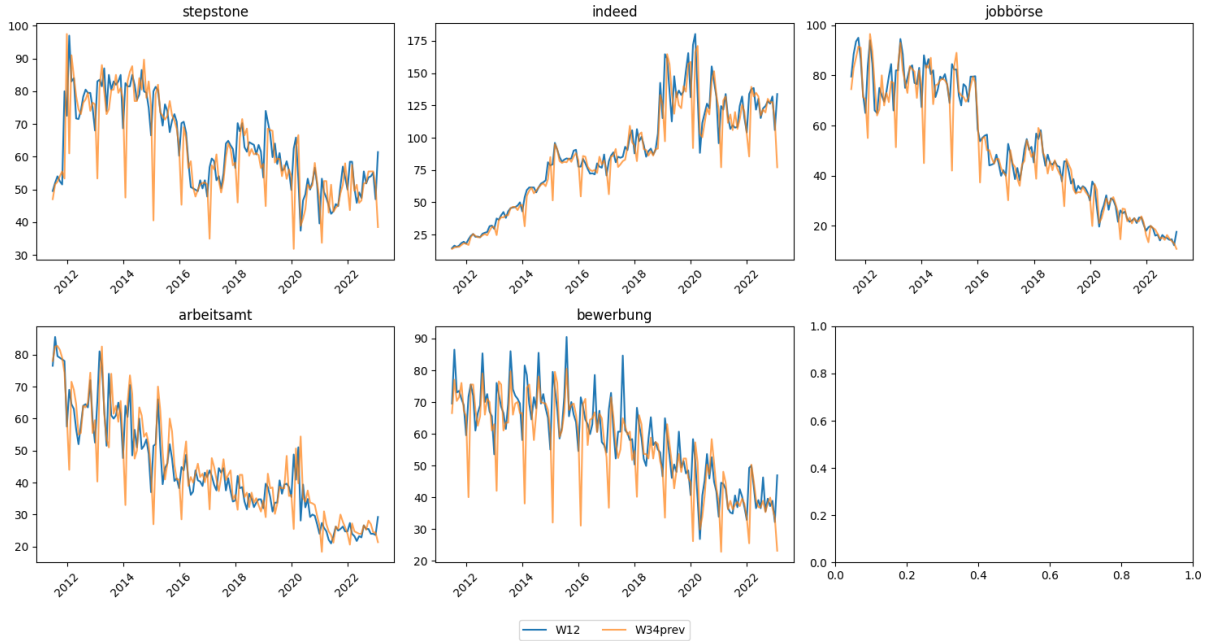


Figure 1: Biweekly Google search indicators for all keywords: W12 (blue) and W34_{prev} (orange), 2011–2023.

A second consistent pattern is that the W34_{prev} component tends to show sharper and more regular seasonal spikes than the corresponding W12 values, particularly for *arbeitsamt* and *bewerbung*. This distinction between the two biweekly components is also

Table 1: Descriptive statistics for biweekly Google Trends indicators and unemployment

Variable	Mean	Std. dev.	Min	Max
stepstone_W12	63.68	13.00	37.36	97.00
indeed_W12	87.77	39.97	14.50	180.24
jobbörse_W12	52.13	24.17	12.24	95.00
arbeitsamt_W12	43.89	15.67	21.02	85.50
bewerbung_W12	58.56	13.83	26.80	90.50
stepstone_W34prev	61.71	14.01	31.81	97.50
indeed_W34prev	84.64	39.47	14.00	170.95
jobbörse_W34prev	50.27	23.82	10.80	96.50
arbeitsamt_W34prev	44.30	16.02	18.39	82.67
bewerbung_W34prev	55.66	14.36	22.78	80.50
Unemployment	6.00	0.72	4.80	7.40

clearly visible in the seasonal boxplots in Figures 3 and 4. For the W34_{prev} indicators, the January spike is particularly pronounced and occurs consistently across all keywords, followed by a sharp decline into February. This pattern is much less pronounced in the

W12 components, where month-to-month differences are visibly flatter.

Long-run patterns differ systematically across keywords. Searches for *arbeitsamt* and *jobbörse* display a persistent downward trend over 2011–2023, consistent with declining usage of public employment-service channels. By contrast, private job portals such as *indeed* and *stepstone* exhibit a pronounced upward trajectory during the 2010s, reflecting structural digitalisation and the increased importance of private platforms in German job search. The keyword *bewerbung* exhibits a smooth and monotonic decline, consistent with reduced reliance on generic application-related searches in favour of platform-integrated application processes.

Several series also feature episodic bursts, most notably around 2020–2021 during the COVID-19 shock. Spikes in *arbeitsamt*, *indeed* and *stepstone* suggest heightened job-search uncertainty and sudden shifts in labour-market activity during the pandemic. In contrast, *jobbörse* and *bewerbung* appear less affected, consistent with their declining long-term usage. Overall, the biweekly aggregation appears internally coherent: for each keyword, the W12 and W34_{prev} components track similar underlying behavioural patterns, with W34_{prev} often exhibiting a clearer and more regular seasonal structure.

The stitched weekly series form the basis for the biweekly aggregation described above and for the unrestricted MIDAS models. Visual inspection of the weekly indices in Figure 2 confirms that they exhibit the same broad seasonal patterns, long-run trends and COVID-related spikes as their biweekly counterparts, with additional high-frequency variation at the weekly level. For brevity, I focus on the biweekly indicators in the main descriptive analysis and use the weekly series directly only in the mixed-frequency specifications.

4 Stationarity, differencing and implications for model choice

Because the subsequent AR(1), ARX and MIDAS specifications are formulated under the assumption of (approximate) covariance stationarity, it is important to assess whether the monthly unemployment rate and the biweekly Google Trends indicators are stationary or contain a unit root. This determines whether the models should be estimated in levels or in (seasonal) differences. In contrast to Askitas and Zimmermann (2009), who estimate their error-correction model directly in seasonal differences without reporting formal unit-root diagnostics, I explicitly examine the integration properties of unemployment and the search indicators before choosing the appropriate specification.

To this end, I apply Augmented Dickey–Fuller (ADF) tests to all series in levels and, where appropriate, in first differences. The ADF test extends the classical Dickey–Fuller regression by adding lagged differences to control for serial correlation in the residuals. For each series, the number of lagged differences is selected automatically using the Akaike

Series	Det. spec	Lags	p-value	Reject H_0 at 5%?
stepstone_W12	c	13	0.4368	No
indeed_W12	ct	13	0.4678	No
jobbörse_W12	ct	12	0.0950	No
arbeitsamt_W12	ct	11	0.7232	No
bewerbung_W12	ct	12	0.5619	No
stepstone_W34prev	c	13	0.5740	No
indeed_W34prev	ct	13	0.5983	No
jobbörse_W34prev	ct	12	0.1206	No
arbeitsamt_W34prev	ct	13	0.4282	No
bewerbung_W34prev	ct	12	0.6317	No
Unemp	c	13	0.3859	No

information criterion (AIC), subject to a maximum of 13 lags, and critical values are taken from MacKinnon (2010) as implemented in the `statsmodels` package.

Because the ADF test is sensitive to the choice of deterministic terms, I select the appropriate specification (no constant, constant, or constant and linear trend) based on an inspection of the plotted series. Series that exhibit a clear trend in levels, such as the unemployment rate and most Google search indices, are tested under the trend specification, while series fluctuating around a nonzero mean are tested with a constant only. This preserves power against the relevant alternatives. Following Hamilton (1994, Table 17.3), the Augmented Dickey–Fuller regression with p augmenting lags can be written in its most general form as

$$y_t = \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \cdots + \zeta_{p-1} \Delta y_{t-p+1} + \rho y_{t-1} + \alpha + \delta t + \epsilon_t \quad (2)$$

where α is an intercept, δt a deterministic linear trend, and the lagged differences Δy_{t-j} control for serial correlation in the innovations. The standard ADF test for a unit root corresponds to testing $H_0: \rho = 0$ against $H_1: \rho < 0$, with different deterministic cases obtained by imposing $\alpha = 0$ and/or $\delta = 0$. The explicit formulations of cases 1, 2 and 4 are summarised in Appendix 11.3.

For each series, I estimate (2) in levels under the appropriate deterministic specification and select the number of augmenting lags p by minimising the Akaike information criterion (AIC) over lag orders from 0 to 13, as implemented in the `statsmodels` package. Table ?? reports the ADF results for the variables in levels. Across all Google Trends indicators and for the unemployment rate, the null hypothesis of a unit root cannot be rejected at the 5% significance level, indicating that the series are nonstationary in levels. To confirm whether the variables are integrated of order one, I repeat the tests for the first differences. As shown in Table 3, the ADF statistics for the first-differenced series strongly reject the unit-root null for all variables. Taken together, the results imply that all monthly variables are $I(1)$ and become stationary after first differencing.

For the high-frequency MIDAS models, the variables enter at their original weekly frequency rather than in biweekly averages. I therefore also examine the stationarity properties of the weekly Google Trends indicators. The ADF results reported in Table 4 show that the weekly *stepstone* and *indeed* series remain nonstationary in levels, whereas the weekly *arbeitsamt*, *jobbörse* and *bewerbung* series reject the unit-root null under the trend specification and thus behave as trend-stationary processes. Overall, the stationarity analysis establishes a clear pattern for the modelling strategy adopted in the remainder of the paper. For the biweekly ARX models, all Google Trends indicators and unemployment are treated as $I(1)$ variables and therefore enter in first differences unless cointegration can be shown to hold. For the unrestricted MIDAS regressions, the weekly indicators provide a richer frequency structure: trend-stationary weekly series such as *jobbörse*, *arbeitsamt* and *bewerbung* can be included in levels while controlling for seasonality, other series such as *stepstone* and *indeed* are entered in first differences. These properties ensure that the subsequent forecasting models rest on well-defined time-series behaviour and avoid spurious regressions arising from nonstationary combinations of unrelated series.

4.1 A notion on cointegration

Since the unemployment rate and the biweekly Google Trends indicators are found to be $I(1)$, one possibility would be to estimate an Error Correction Model (ECM) if the variables share a stable long-run relationship. In this context it is important to note that the ECM specification used by Askitas and Zimmermann (2009) is not derived from a formal cointegration analysis (Engle and Granger, 1987). Neither study performs an Engle–Granger nor a Johansen test, and the “error-correction term” in their model is implicitly taken to be the lagged level of unemployment rather than the residual from a cointegrating regression. Consequently, the specification resembles an autoregressive model in differences with seasonal controls rather than a structurally motivated ECM.

To assess whether a genuine long-run relationship exists in the present data, I test for pairwise cointegration between unemployment and each biweekly Google Trends indicator using the Engle–Granger procedure. In the first step, unemployment is regressed on each search indicator in levels (including a constant and seasonal dummies), and the residuals are extracted. Under cointegration, these residuals should be stationary. The appropriate critical values are those of Case 2 in Hamilton (1994), since the long-run regression includes a constant but no trend.

Before turning to the test statistics, it is useful to note that several Google Trends series, most notably *indeed* and *stepstone*, exhibit pronounced upward trends that are not mirrored in the unemployment series. From an economic perspective, this already weakens the case for a stable long-run equilibrium relationship. The formal Engle–Granger tests confirm this intuition. Table 6 reports the Engle–Granger ADF statistics for the residuals of the

long-run regressions between unemployment and each biweekly Google Trends indicator. Using the appropriate Case 2 critical values from Hamilton (1994), the unit-root null is rejected at the 5% level only for two series (`stepstone_W12` and `indeed_W34prev`), while all remaining eight indicators clearly fail to reject the null. Even for the two borderline cases, the test statistics lie only slightly below the 5% threshold, and the rejection is unstable across deterministic specifications.

The statistical evidence is further undermined by the visual behaviour of the Engle–Granger residuals (Figure 5). Rather than exhibiting stationary mean-reverting dynamics, the residuals display long, persistent drifts, seasonal patterns, structural breaks, and pronounced deviations during episodes such as the COVID–19 crisis. Such behaviour is inconsistent with the notion of a stable equilibrium error term and therefore contradicts the existence of a meaningful cointegrating relationship.

Taken together, these results imply that the unemployment rate and the biweekly search indicators are not cointegrated. In the absence of cointegration, an ECM is not appropriate, and following the Granger Representation Theorem, the correct empirical specification for non-cointegrated $I(1)$ variables is an ARX model in first differences. This avoids spurious regression behaviour and ensures that the short-run dynamic effects of search activity on unemployment are identified without imposing an unfounded long-run equilibrium structure.

Cointegration is not assessed for the weekly Google Trends indicators used in the subsequent MIDAS models, because the MIDAS framework is designed to capture short-run mixed-frequency dynamics rather than long-run equilibria. Weekly variables therefore enter the MIDAS regressions either in levels (when adequately detrended) or in first differences, depending on their stationarity properties.

5 Econometric Models

The empirical analysis adopts a nowcasting perspective. Nowcasting refers to the real-time prediction of the present or very recent past, i.e. producing an estimate of the unemployment rate for month M before the official figure is released. At that point, the forecaster observes unemployment only up to $M - 1$, while weekly Google search indicators are already available within month M . The econometric task is therefore to map past unemployment and high-frequency search information into a real-time estimate of U_M .

To address this task, I consider three classes of models. As a benchmark, unemployment is modelled as a univariate autoregressive process in first differences. Second, ARX models augment this benchmark with biweekly Google Trends indicators in order to assess whether search activity improves the nowcast relative to using unemployment alone and to compare the information content of early versus late within-month search signals. Third, an unrestricted MIDAS regression incorporates weekly Google Trends series directly into

the monthly equation, allowing the mixed-frequency data to be exploited without temporal aggregation. The formal specification of each model is given below.

5.1 AR(1)

Since unemployment was found to be non-stationary in levels (Table ??) but stationary in first differences, the nowcasting target variable is the monthly change in unemployment, ΔU_t the corresponding AR(1) benchmark model is given by

$$\Delta U_t = \phi \Delta U_{t-1} + \epsilon_t \quad (3)$$

This specification models the monthly change in unemployment as a function of its own lag and serves as a simple benchmark that relies solely on past unemployment information.

5.2 ARX

To evaluate whether Google search activity contains information beyond past unemployment, I augment the AR(1) benchmark with biweekly Google Trends indicators. The resulting ARX model is

$$\Delta U_t = \alpha + \phi \Delta U_{t-1} + \sum_{i=1}^K \sum_{\ell=0}^{L_i} \beta_{i,\ell} \Delta G_{i,t-\ell} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t, \quad (4)$$

where $\Delta G_{i,t-\ell}$ denotes the lagged changes of the i -th Google Trends series and $D_{m,t}$ are monthly dummy variables capturing residual seasonality in the non-seasonally adjusted unemployment data.

In this specification, the monthly change in unemployment, ΔU_t , depends on its own lag, on lagged changes in the high-frequency search indicators, and on seasonal controls. The term ΔU_{t-1} captures persistence in unemployment changes, while the biweekly Google Trends terms allow the model to exploit information arriving within the month.

Since unemployment in levels is integrated of order one, all models are estimated in first differences. Working with ΔU_t and $\Delta G_{i,t}$ ensures stationarity of the regression and avoids spurious relationships, while the AR(1) structure provides a parsimonious benchmark for monthly unemployment dynamics.

5.3 MIDAS

Mixed-data sampling (MIDAS) regressions, originally introduced by Ghysels et al. (2004, 2006), provide a flexible econometric framework for modelling relationships between variables observed at different sampling frequencies. A central motivation for MIDAS arises in situations where a low-frequency target series (e.g., monthly or quarterly macroeconomic

indicators) is potentially explained by high-frequency predictors (such as daily financial variables, weekly surveys, or—in the present case—weekly Google search activity). MIDAS regressions allow the high-frequency information to enter directly into the low-frequency equation without requiring temporal aggregation, interpolation, or large state-space systems. This property has made MIDAS models widely used in empirical finance (e.g., volatility forecasting, return predictability) and in macroeconomic nowcasting, where timely indicators are essential.

The original MIDAS formulations proposed rely on parametric weighting schemes, most prominently the Beta and Almon lag polynomials, to summarise a potentially large number of high-frequency lags in a parsimonious way. These so-called restricted MIDAS models impose a functional form on how weekly or daily lags of the predictor affect the low-frequency outcome. The weighting function is governed by only a few shape parameters, which keeps estimation tractable even when the frequency ratio between the series is large. Other extensions of the MIDAS framework, such as transfer-function MIDAS (TF-MIDAS), which incorporates ARMA-type dynamics in the weighting process, have been proposed in the literature (Bonino-Gayoso and Garcia-Hiernaux, 2021), but these approaches typically involve more complex estimation procedures and are therefore beyond the scope of the present analysis.

A later development is the unrestricted MIDAS (U-MIDAS) model, in which no parametric weighting structure is imposed and each high-frequency lag receives its own coefficient. This variant, discussed for example in Costa et al. (2024), is estimated via ordinary least squares and avoids potential misspecification stemming from an incorrect weighting function. U-MIDAS is therefore attractive whenever full flexibility in capturing the contribution of individual weekly lags is preferred, or when the number of available high-frequency observations is sufficiently modest to allow estimation without overparameterisation.

In the context of this analysis, the unrestricted specification is particularly suitable for two reasons. First, the frequency ratio between weekly Google Trends and monthly unemployment is relatively small, making an unrestricted lag structure feasible. Second, my primary interest lies in assessing whether weekly search activity contains predictive power for unemployment, rather than in estimating a particular weighting function. I therefore adopt the U-MIDAS regression as the mixed-frequency counterpart to the biweekly ARX models introduced above.

The unrestricted MIDAS (U-MIDAS) regression allows each weekly lag of the high-frequency Google Trends series to enter the monthly unemployment equation with its own coefficient. The model is specified as

$$\Delta U_t = \alpha + \sum_{k=0}^K \beta_k \Delta G_{t-k/m} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t, \quad (5)$$

where $\Delta G_{t-k/m}$ denotes the k -th weekly lag of the differenced Google Trends indicator

and $D_{m,t}$ are monthly dummy variables controlling for residual seasonality. Unlike the restricted MIDAS models of Ghysels et al. (2004, 2006), the U-MIDAS specification imposes no parametric weighting scheme on the high-frequency lags. This has the important advantage that the model can be estimated by ordinary least squares (OLS), as discussed in Foroni et al. (2013). For the present application, OLS estimation is particularly attractive because it avoids the nonlinear optimisation required in restricted MIDAS models, facilitates repeated estimation in a rolling-window nowcasting framework, and allows the contribution of individual weekly lags to be determined directly from the data without imposing a specific functional form.

6 Methodology

This section outlines the empirical strategy used to assess whether biweekly and weekly Google search indicators enhance the real-time prediction of monthly German unemployment. All models introduced in Section Econometric models are estimated in a pseudo-real-time environment that replicates the information constraints faced by a forecaster, and their predictive performance is evaluated over a common out-of-sample period.

6.1 Real-time information sets and forecasting design

At the end of month t , the forecaster observes the unemployment rate only up to $t - 1$, because the figure for month t is released with a publication lag despite being measured in the middle of the month. In contrast, the weekly Google Trends indicators are available almost immediately and can therefore be used to form beliefs about the current state of the labour market. Letting $\Delta G_s^{(w)}$ denote the weekly change of a given search indicator, the real-time information set available at the end of month t is

$$\mathcal{I}_t = \{\Delta U_1, \dots, \Delta U_{t-1}; \Delta G_1^{(w)}, \dots, \Delta G_t^{(w)}\}. \quad (6)$$

Each nowcast for month t is generated using only the information contained in $\mathcal{I}_t$. Since unemployment in levels is non-stationary, all models are estimated in first differences to ensure that the forecasting equations are well-behaved. The resulting prediction for the monthly change in unemployment, $\widehat{\Delta U}_t$, is then converted into a level forecast via

$$\widehat{U}_t = U_{t-1} + \widehat{\Delta U}_t. \quad (7)$$

6.2 Rolling estimation window

The models are estimated using a rolling-window framework, which mirrors the updating process of a real-time forecaster. For each prediction date t , the estimation sample consists

of all observations strictly prior to t . The procedure begins once a sufficiently long history of data is available; in the present application, a minimum of 60 monthly observations is required so that lagged terms and seasonal dummies are reliably identified. At each step, the model parameters are re-estimated using the expanding information set, and a one-step-ahead nowcast for month t is produced.

6.3 Evaluation sample and forecast horizon

The evaluation period spans January 2019 to January 2023, reflecting both the availability of weekly Google Trends data and the publication lag of the unemployment series. For every month in this interval, a real-time nowcast $\hat{U}_t$ is produced and compared with the subsequently released unemployment figure U_t . The resulting sequence of forecast errors,

$$e_t = \hat{U}_t - U_t, \quad (8)$$

provides the basis for comparing the predictive performance of the competing models.

6.4 Forecast evaluation models

Forecast accuracy is assessed using root-mean-squared and mean absolute forecast errors:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{U}_t - U_t)^2}, \quad \text{MAE} = \frac{1}{T} \sum_{t=1}^T |\hat{U}_t - U_t| \quad (9)$$

These metrics quantify how well each model predicts the actual unemployment level.

6.5 Statistical comparison of predictive accuracy

Pairwise comparisons of predictive accuracy across models use the Diebold–Mariano (DM) test with squared error loss. The null hypothesis of the DM test is equal predictive accuracy:

$$H_0 : E[d_t] = 0, \quad d_t = e_{1,t}^2 - e_{2,t}^2. \quad (10)$$

Rejections indicate statistically significant improvements of one model over another.

7 Results and Discussion

This section presents the empirical findings and compares the predictive performance of the econometric models introduced above. The analysis proceeds in three steps. First, the benchmark AR(1) model is evaluated against the ARX specifications to assess whether biweekly Google Trends indicators provide incremental predictive power for monthly

unemployment. This comparison allows us to quantify the value of including high-frequency search information relative to a model that relies solely on past unemployment dynamics. Second, the unrestricted MIDAS regression is examined to determine whether weekly Google search activity improves nowcasting performance beyond what can be achieved with biweekly aggregation. Since the MIDAS framework exploits the full weekly resolution of the data, this comparison provides evidence on whether additional high-frequency structure contains useful information for predicting unemployment.

Finally, the two biweekly aggregation schemes are contrasted to investigate whether early-month or late-month search activity carries greater predictive content. This exercise revisits the argument in Askitas and Zimmermann (2009) regarding the timing of online search responses to labour market conditions and evaluates whether that pattern persists over a longer and more recent time period.

7.1 Benchmark result: AR(1) Nowcast

The AR(1) model in first differences serves as the baseline against which the predictive value of Google Trends indicators is evaluated. Across the full evaluation window (January 2019–January 2023), the benchmark yields an RMSE of 0.182 and an MAE of 0.103, indicating that a simple autoregressive structure already captures a sizable part of the month-to-month variation in German unemployment.

Figure 7 displays the absolute nowcast errors over time. The error profile is generally stable and low, with most deviations remaining below 0.2 percentage points. A single pronounced spike occurs in early 2020, coinciding with the onset of the COVID-19 shock. This period is characterised by an abrupt and unprecedented labour-market adjustment, which a univariate autoregressive model cannot anticipate. Apart from this singular episode, only mild error increases appear in mid-2022 and at the end of the sample, but these remain modest relative to the pandemic-related deviation.

A closer inspection of the predicted monthly changes (Figure 7), shows that the AR(1) model reproduces moderate fluctuations reasonably well but systematically struggles around turning points. During the first COVID wave (spring 2020), the model reacts with a noticeable delay and understates the magnitude of the jump in unemployment. Similar, though far smaller, discrepancies arise in late 2021 and mid-2022, where the model tends to either anticipate changes slightly too early or overshoot the observed variation. Outside such episodes, the alignment between actual and predicted changes is relatively tight, particularly in periods of gradual and monotonic labour-market adjustment.

These patterns are consistent with the properties of an AR(1) process in first differences: shocks propagate smoothly, and predictions depend solely on the recent past of unemployment. When the data-generating process is subject to large, abrupt disturbances or short-lived policy effects, the benchmark lacks the information required to react contem-

poraneously. The evidence therefore highlights both the strength and the limitation of the AR benchmark: it performs reliably in stable periods but fails precisely when timely high-frequency information would be expected to deliver gains. In the next section we will see whether added Google search activity can improve real-time prediction, particularly around turning points where the AR(1) model exhibits its most visible weaknesses.

7.2 Introduction of Google search terms

The ARX specification extends the autoregressive benchmark by augmenting unemployment changes with biweekly Google Trends indicators and seasonal dummies. This design aims to evaluate whether search activity provides predictive information beyond the autoregressive component, particularly around turning points where the AR(1) model exhibits its largest deviations. Across all indicators and both aggregation schemes (W12 and W34prev), forecast accuracy remains tightly clustered around the benchmark. The best-performing W12 indicator (jobbörse in Week W12) achieves an RMSE of 0.180 and an MAE of 0.102, whereas the best W34prev indicator (arbeitsamt in the previous weeks 3 and 4) yields an RMSE of 0.182 and an MAE of 0.106. These values differ from the AR(1) benchmark by less than one percent in either direction, indicating that the inclusion of Google Trends variables does not materially improve predictive performance.

Figure 8 illustrates the forecast paths for the two best-performing specifications. The ARX trajectories essentially mirror those of the AR(1) model: unemployment levels are tracked closely during periods of gradual adjustment, but the models respond sluggishly to sudden shifts. During the onset of the COVID-19 shock in early 2020, both ARX specifications underestimate the magnitude and timing of the labour-market deterioration. The forecasts begin to adjust only after unemployment has already risen sharply, reflecting the dominance of the autoregressive term and the limited contemporaneous informativeness of the search indicators. A similar pattern emerges around mid-2022, where the ARX models briefly diverge from observed unemployment, either declining too quickly or adjusting with a short delay.

The absolute forecast errors, shown in Figure 9, reinforce this interpretation. The temporal error profile is nearly identical to that of the AR benchmark: errors remain small throughout most of the sample, with a single, dominant spike between March and June 2020 and smaller increases in January 2021 and July 2022. The fact that neither the W12 nor W34prev indicators attenuate these deviations suggests that weekly search intensity does not deliver timely signals of abrupt changes in unemployment. Instead, the ARX models inherit the same weaknesses as the AR(1) structure, reacting only once the shift is visible in the lagged unemployment data.

A decomposition of forecast accuracy by calendar month further highlights the limited incremental value of Google Trends. For both variables, the RMSE peaks consistently in

April, May, and June, precisely the months with the highest AR(1) errors, while forecast errors fall markedly between September and December. The near-identical seasonal pattern across all ARX models indicates that the monthly dummy variables, rather than the search indicators, drive improvements in model fit. In particular, the Google Trends coefficients do not meaningfully alter the seasonal error structure or reduce uncertainty in months associated with systematic labour-market transitions.

The analysis of predicted versus actual monthly unemployment changes (Figure 10) provides a final piece of evidence. While the ARX models capture moderate fluctuations in ΔU_t , their reactions remain visibly lagged around major turning points. The COVID spike is predicted too late and too weakly, and the dynamics surrounding the mid-2022 decline exhibit exaggerated responses not reflected in the data. These patterns demonstrate that the high-frequency nature of Google Trends does not translate into leading information at the monthly horizon once the autoregressive component is taken into account.

Overall, the evidence shows that biweekly search intensity measures have minimal incremental predictive value for monthly unemployment. Although some indicators perform slightly better than others, the gains are economically negligible and do not alter the underlying forecast dynamics. The ARX models remain dominated by the autoregressive structure, and the Google Trends series do not provide the forward-looking content necessary to improve forecasts at precisely the moments when timely information would be most valuable. The next section examines whether weekly MIDAS regressions, are better suited to extract predictive signals from search behaviour.

7.3 Unrestricted MIDAS results

The unrestricted MIDAS (U-MIDAS) framework exploits the full weekly resolution of Google search activity by stacking eight weekly differences of a single search term as regressors for the monthly unemployment change. Unlike the ARX specification, which compresses high-frequency information into biweekly aggregates, the MIDAS design allows for heterogeneous short-run responses across the weeks preceding each monthly release. This section evaluates whether such finer temporal structure yields incremental predictive value.

Across all five weekly Google Trends series, forecast accuracy improves only marginally relative to the AR(1) and ARX benchmarks. Table 7 reports the nowcast performance for each indicator. The best-performing MIDAS model uses jobbörse, achieving an RMSE of 0.172 and an MAE of 0.096, while stepstone performs slightly worse (RMSE 0.174). The remaining indicators exhibit RMSE values between 0.185 and 0.190. Although the best MIDAS specification outperforms the ARX benchmark (0.180) by roughly 0.5–1 percentage points, the improvement is economically modest and well within the range of sampling variation.

Figure 11 displays the level predictions for the best MIDAS model. As in the ARX and AR(1) frameworks, the MIDAS trajectory tracks unemployment closely during periods of gradual adjustment but fails to anticipate sudden turning points. The most pronounced miss occurs during the first COVID-19 shock between March and May 2020, where the MIDAS nowcast underestimates unemployment by up to 0.92 percentage points, the worst deviation in the entire sample. Smaller but systematic deviations reappear in June 2022, a period also associated with structural shifts in labour-market conditions that high-frequency search data do not seem to capture timely.

The absolute forecast error series in Figure 12 confirms this pattern. Errors remain low and relatively stable in normal times, but spike sharply during crisis episodes, mirroring the behaviour observed in both AR(1) and ARX forecasts. Weekly search intensity therefore does not generate forward-looking information capable of mitigating the model’s shortcomings precisely when predictions are most challenging.

The decomposition of predictive accuracy by month (Figure 13) reveals yet another parallel to the ARX results. Forecast errors peak consistently in April and May, with RMSE values exceeding 0.45, before declining steadily through the summer and reaching minima in September and October (0.05–0.10). The near-identical seasonal pattern across AR(1), ARX, and MIDAS specifications indicates that the monthly dummy variables, rather than the high-frequency regressors, account for most improvements in fit. This suggests that the weekly search terms do not contribute meaningful seasonal structure beyond what is already absorbed by deterministic monthly effects.

Turning to the dynamics of predicted vs actual monthly changes in unemployment (Figure 14), the MIDAS model captures moderate fluctuations but remains visibly delayed around major movements. The 2020 spike is again predicted too weakly and too late; similarly, the model overreacts during mid-2022, producing a downward adjustment that substantially exceeds the realised change. These discrepancies highlight the dominance of the autoregressive component in the estimated coefficients remains highly significant—while only a minority of the weekly lags exhibit statistical significance, and even then with small magnitudes. In effect, the high-frequency regressors do not materially alter the short-run propagation mechanism of unemployment.

The overall evidence therefore points to a clear conclusion: while unrestricted MIDAS regressions yield slightly lower RMSEs than biweekly ARX models, the improvements are small, unstable across indicators, and do not address the model’s structural weaknesses at turning points. Weekly search activity, even when exploited at its full temporal resolution, does not provide consistent leading signals for monthly unemployment in Germany.

Figure 15 summarises the real-time nowcasts from the three main specifications. All models track the level of unemployment closely over the evaluation window, and the forecast paths are visually almost indistinguishable during periods of gradual adjustment. The unrestricted MIDAS specification yields slightly lower errors in tranquil phases, whereas the

AR(1) and ARX models occasionally overshoot or undershoot by a small margin. However, none of the models succeeds in anticipating the sharp movements around the onset of the COVID-19 shock or the mid-2022 reversal; instead, all three paths adjust only after the turning points have already occurred. This visual evidence already suggests that the incremental information contained in Google search activity is modest and that the autoregressive component dominates the forecasting dynamics.

7.4 Evaluation W12 vs W34 comparison

A central empirical claim in Askitas and Zimmermann (2009) is that the timing of Google search activity within a month matters for unemployment prediction. Their argument is that searches conducted early in the month (weeks 1–2, here W12) are less contaminated by contemporaneous labour-market news and may therefore contain more forward-looking information than late-month searches (weeks 3–4, here W34prev). To evaluate this hypothesis, both aggregation schemes were benchmarked against each other for all five keywords used in the analysis. Table 8 reports RMSEs for W12 and W34prev along with Diebold–Mariano (DM) tests of equal predictive accuracy. Three findings stand out clearly. First, forecast accuracy is extremely similar across the two aggregation schemes. For every keyword, the RMSE difference between W12 and W34prev is below 0.01, and in absolute terms the two specifications produce nearly indistinguishable forecast paths. The smallest difference appears for *arbeitsamt*, where W34prev performs marginally better (RMSE 0.182 vs. 0.184), while for *stepstone* and *jobbörse* W12 is slightly more accurate. These differences are economically negligible.

Second, the Diebold–Mariano tests uniformly fail to reject equal predictive accuracy. All DM statistics lie well within plus and minus 1.3, and all p-values exceed 0.20, with most above 0.40 or 0.80. Because the DM test is based on the time series of squared forecast errors, a statistically significant difference would require consistent superiority across the full evaluation period. No such effect appears in the data.

Third, the absence of detectable timing effects is consistent with the earlier ARX diagnostics. Search behaviour moves broadly in line with unemployment itself and does not exhibit systematic informational advantages at either the beginning or end of the month. The autoregressive term dominates the forecasting equation, and once seasonality is controlled for, weekly search intensities contribute little incremental predictive signal.

Taken together, the results indicate that the timing channel proposed by Askitas and Zimmermann (2009) does not appear in the German unemployment data for the 2011–2023 period. Neither early-month nor late-month search activity produces systematically lower forecast errors, and the differences are statistically indistinguishable from noise. Consequently, the choice between W12 and W34prev can be viewed as empirically irrelevant in this setting, and the remainder of the analysis focuses on whether retaining the full

weekly structure, rather than aggregating to biweekly series, yields more informative predictors.

7.5 Discussion

The empirical results demonstrate a remarkably consistent pattern across all model classes: neither biweekly nor weekly Google search indicators provide substantial incremental information for predicting monthly unemployment in Germany. While this finding contrasts with some early studies that reported sizeable gains from high-frequency search activity, it is fully plausible once statistical, economic, and historical considerations are taken into account.

From a statistical perspective, the behaviour of German unemployment over the sample period is dominated by strong persistence and seasonality. The AR(1) benchmark already captures the majority of month-to-month variation, and the inclusion of deterministic monthly effects absorbs most of the recurring seasonal structure. Once these components are accounted for, little residual variation remains for high-frequency regressors to explain. The MIDAS and ARX estimates confirm this: the autoregressive term consistently displays the highest significance and largest magnitude, whereas Google Trends coefficients are small, unstable, and rarely significant. In essence, the high-frequency series do not alter the fundamental propagation mechanism of unemployment, and their marginal explanatory power is overshadowed by the smooth autoregressive dynamics.

Economically, the limited value of online search activity likely reflects the institutional features of the German labour market. Job loss and unemployment registration are mediated through formal and administrative processes that evolve slowly relative to weekly search intensity. Employers' hiring decisions, social insurance mechanisms, and short-time work arrangements (*Kurzarbeit*) introduce inertia into labour-market transitions, producing adjustments that unfold over months rather than weeks. Because layoffs and job-search behaviour often coincide with, rather than precede, observable changes in unemployment, search intensity may track current labour-market conditions but does not provide a strong leading signal. The timing results, which show no systematic advantage of early- or late-month search queries, reinforce this interpretation.

Historical developments during the sample further limit the potential usefulness of Google Trends indicators. The most pronounced forecast errors occur during the COVID-19 shock, an event characterised by abrupt and policy-driven labour-market adjustments. The unprecedented expansion of *Kurzarbeit*, rapid policy responses, and temporary shutdowns disrupted normal job-search behaviour. During this period, search intensity became decoupled from realised unemployment: many individuals were placed on short-time work rather than entering unemployment, while others delayed job search amid uncertainty. As a result, neither biweekly nor weekly search trends provided early warnings of the

labour-market disruption, and models responded only after the shock had materialised in the administrative unemployment data. Even outside the pandemic, changes in online job-search behaviour—such as the increasing role of mobile platforms and professional networks—may weaken the link between specific Google keywords and labour-market transitions.

Taken together, these observations suggest that the forecasting environment is one in which autoregressive structure, institutional inertia, and deterministic seasonality outweigh any short-term fluctuations in search intensity. The smooth and persistent nature of unemployment dynamics limits the potential benefits of including noisy, volatile weekly indicators. Consequently, the MIDAS and ARX models make only marginal improvements over the AR(1) benchmark and fail precisely at the turning points where high-frequency data might be expected to add value.

8 Conclusion

This paper evaluated whether high-frequency Google search activity contains useful predictive information for nowcasting monthly unemployment in Germany. Using biweekly ARX models and unrestricted MIDAS regressions, the analysis assessed three central questions: (i) whether Google Trends improve upon a simple autoregressive benchmark, (ii) whether exploiting the full weekly resolution enhances predictive accuracy, and (iii) whether the timing of search activity within a month matters for forecasting performance.

Across all specifications, the findings point to a clear and consistent conclusion. The autoregressive benchmark already captures most of the month-to-month variation in unemployment, and neither biweekly nor weekly search intensity leads to meaningful improvements in forecast accuracy. Although some models—most notably the MIDAS specification using *jobbörse*—achieve slightly lower RMSEs, the gains are economically small and statistically fragile. Moreover, the forecast paths of all models remain nearly indistinguishable during periods of gradual labour-market adjustment and fail to anticipate the major turning points in the sample, particularly the COVID-19 shock and the mid-2022 reversal. High-frequency search data do not provide reliable early signals of these disruptions; instead, the models adjust only after shifts are visible in lagged unemployment itself.

The comparison of early-month (W12) and late-month (W34prev) aggregation schemes—central to the original argument in Askitas and Zimmermann (2009)—further reveals no systematic timing advantage. RMSE differences are negligible, and Diebold–Mariano tests uniformly fail to reject equal predictive accuracy. This suggests that the timing channel identified in earlier studies does not generalise to the extended 2011–2023 period, likely reflecting both structural changes in job-search behaviour and the stabilising effect of institutional labour-market mechanisms.

Overall, the evidence indicates that German unemployment dynamics are shaped predominantly by smooth autoregressive structure and seasonal patterns, leaving little room for noisy high-frequency indicators to contribute additional predictive content. The institutional inertia of the labour market, combined with historical events such as the unprecedented expansion of short-time work during the pandemic, further weakens the contemporaneous link between search intensity and realised unemployment. For real-time labour-market monitoring, these findings suggest caution in relying on Google Trends as leading indicators: while useful for capturing general public attention or behavioural shifts, they do not provide robust short-term forecasting improvements at the monthly frequency. Future research may explore whether alternative forms of online behavioural data, nonlinear machine-learning models, or disaggregated regional unemployment series offer richer predictive content. However, within the framework studied here, the predictive limits of search data appear binding, and traditional autoregressive dynamics remain the dominant driver of accurate nowcasting performance.

9 Tables

9.1 Google Trend variables

Table 2: Comparison of Google Trends variables: Askitas & Zimmermann (2009) vs. this study

Category	Askitas & Zimmermann (2009)	This seminar (2011–2023)
Administrative contact	<i>arbeitsamt, arbeitsagentur</i>	<i>arbeitsamt</i>
Interest in unemployment news	<i>arbeitslosenquote</i>	— (excluded)
Personnel consulting	<i>personalberater, personalberatung</i>	— (excluded)
Job board usage	<i>stepstone, jobworld, jobscout, meinestadt, monster, jobbörse</i>	<i>jobbörse, stepstone, indeed</i>
Application activity	—	<i>bewerbung</i>

9.2 First differences tables

Table 3: Augmented Dickey–Fuller tests on first differences

Series	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_W12	11	-3.2242	-1.9433	0.0013	Yes
indeed_W12	12	-2.4965	-1.9433	0.0121	Yes
jobbörse_W12	11	-3.0608	-1.9433	0.0022	Yes
arbeitsamt_W12	10	-5.5334	-1.9433	0.0000	Yes
bewerbung_W12	12	-3.4875	-1.9433	0.0005	Yes
stepstone_W34prev	11	-4.0136	-1.9433	0.0001	Yes
indeed_W34prev	12	-2.2670	-1.9433	0.0225	Yes
jobbörse_W34prev	11	-2.9936	-1.9433	0.0027	Yes
arbeitsamt_W34prev	12	-3.6222	-1.9433	0.0003	Yes
bewerbung_W34prev	11	-5.1217	-1.9433	0.0000	Yes
Unemp	13	-2.8170	-1.9433	0.0047	Yes

Table 4: Augmented Dickey–Fuller tests in levels (weekly aggregated data)

Series	Det. spec	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone	c	12	-2.3279	-2.8664	0.1631	No
indeed	ct	13	-2.5549	-3.4178	0.3011	No
jobbörse	ct	2	-6.5105	-3.4177	0.0000	Yes
arbeitsamt	ct	0	-9.8365	-3.4177	0.0000	Yes
bewerbung	ct	2	-9.1825	-3.4177	0.0000	Yes

9.3 ADF for levels normal series

9.4 ADF first differences MIDAS

Table 5: ADF tests on first differences of weekly indicators (no deterministic terms)

Series	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_weekly_diff1	12	-10.1611	-1.9415	0.0000	Yes
indeed_weekly_diff1	12	-10.8664	-1.9415	0.0000	Yes

9.5 cointegration results

Table 6: Engle–Granger ADF tests for cointegration (residual-based ADF, Case 2 critical values)

Series	ADF statistic	5% crit. value	p-value	Lags	Obs	Reject H_0 at 5%?
stepstone_W12	-3.6274	-3.4459	0.0276	13	126	Yes
indeed_W12	-3.0409	-3.4456	0.1210	12	127	No
jobbörse_W12	-2.7996	-3.4456	0.1971	12	127	No
arbeitsamt_W12	-3.1826	-3.4459	0.0879	13	126	No
bewerbung_W12	-2.3753	-3.4456	0.3928	12	127	No
stepstone_W34prev	-3.2715	-3.4456	0.0711	12	127	No
indeed_W34prev	-3.6061	-3.4459	0.0293	13	126	Yes
jobbörse_W34prev	-3.1639	-3.4456	0.0918	12	127	No
arbeitsamt_W34prev	-3.0760	-3.4459	0.1121	13	126	No
bewerbung_W34prev	-2.6002	-3.4456	0.2798	12	127	No

9.6 MIDAS models

9.7 Early vs late comparisons

Table 7: Unrestricted MIDAS nowcast performance by weekly Google Trends indicator

Google Trends indicator (weekly)	RMSE	MAE
jobbörse	0.172	0.096
stepstone	0.174	0.102
bewerbung	0.185	0.112
indeed	0.188	0.128
arbeitsamt	0.190	0.107

Table 8: Comparison of W12 and W34prev biweekly aggregation schemes

Keyword	RMSE (W12)	RMSE (W34prev)	DM statistic	p-value
stepstone	0.1837	0.1879	-1.278	0.207
indeed	0.1861	0.1922	-0.719	0.475
arbeitsamt	0.1844	0.1817	0.708	0.483
jobbörse	0.1802	0.1818	-0.882	0.382
bewerbung	0.1840	0.1847	-0.165	0.870

10 figures

10.1 weekly vars

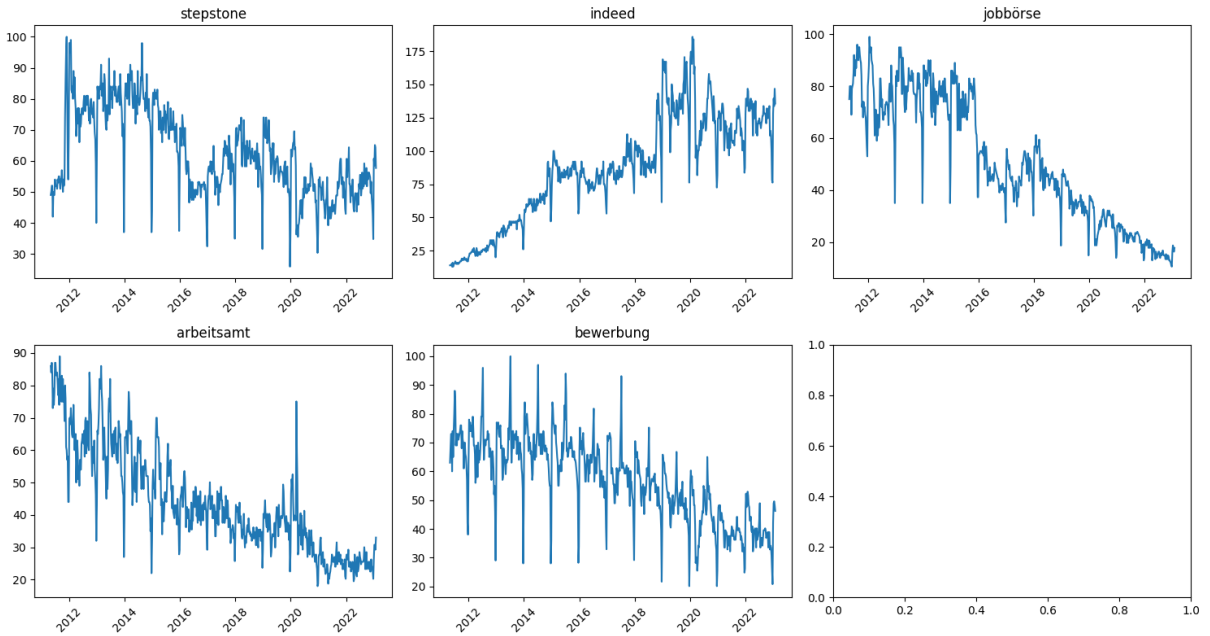


Figure 2: Weekly variables displayed

10.2 Seasonal Boxplot w34prev

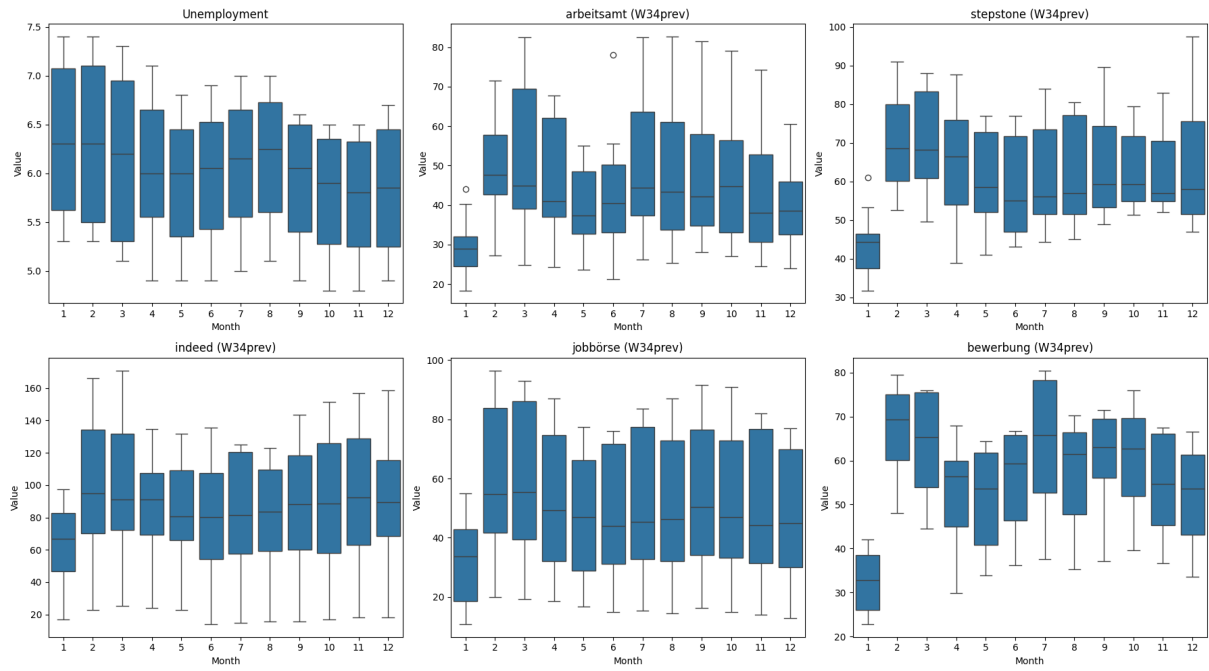


Figure 3: Seasonal boxplots for W34prev indicators.

10.3 seasonal boxplot w12

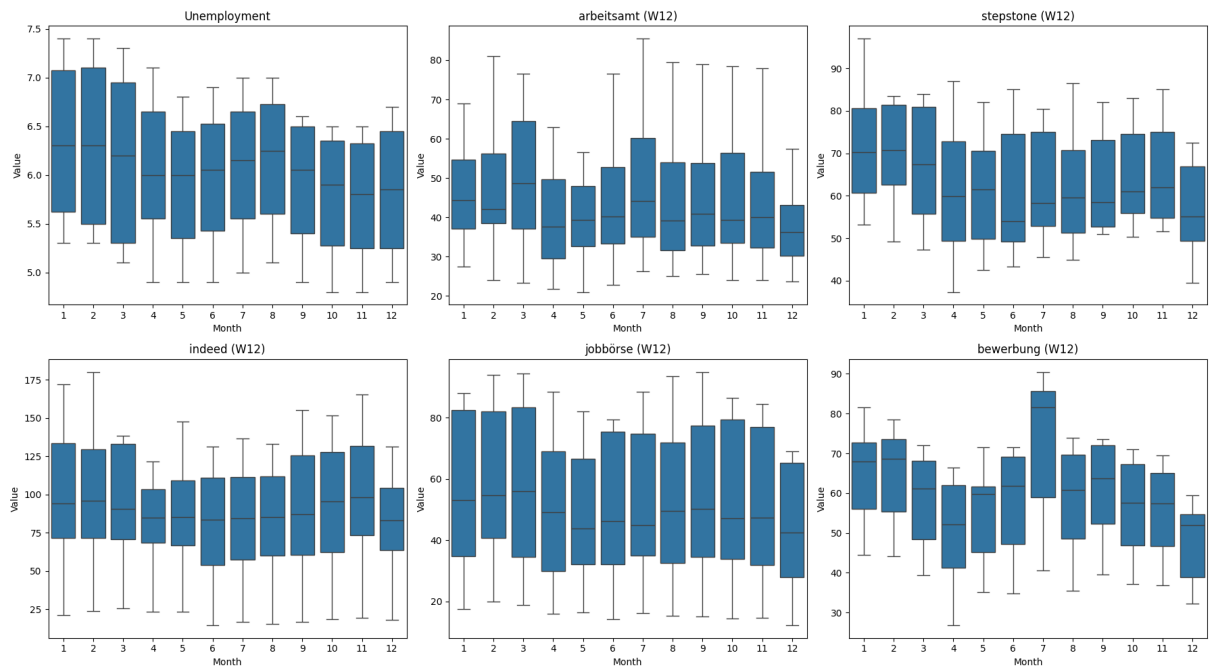


Figure 4: Seasonal boxplots for W12 indicators.

10.4 residuals cointegration

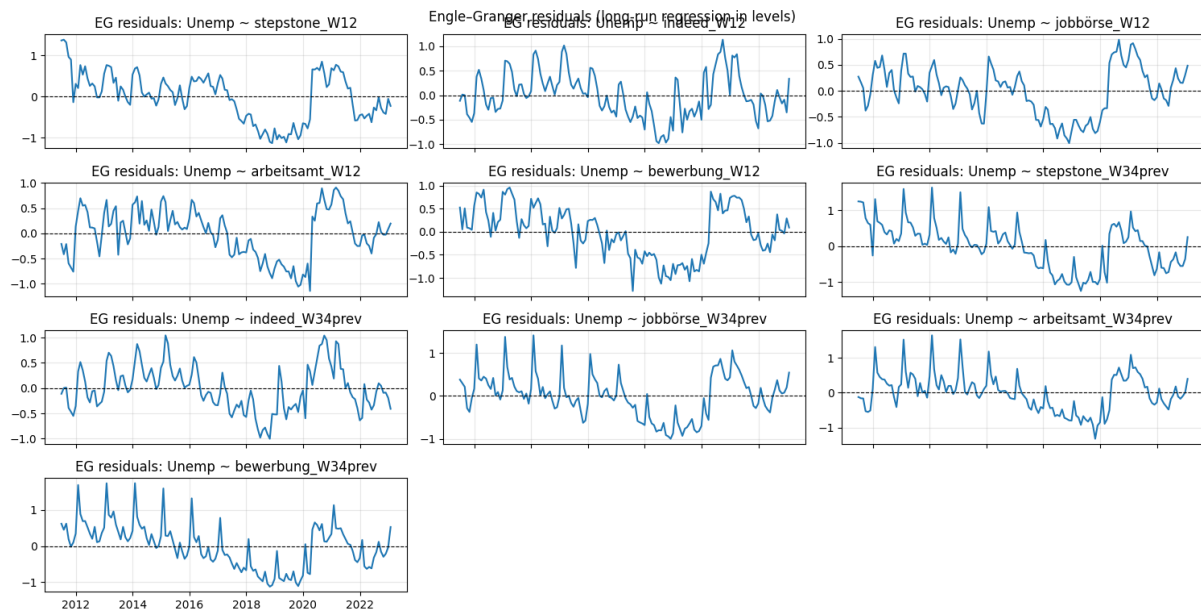


Figure 5: Residuals for cointegration

10.5 AR(1)- Absolute nowcast errors over time

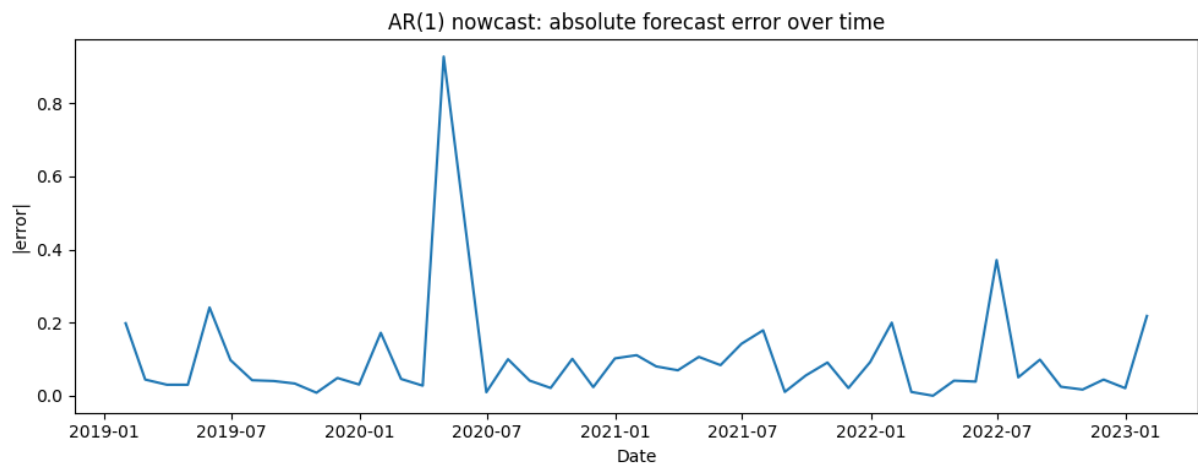


Figure 6: Residuals for cointegration

10.6 AR(1)-predicted monthly changes

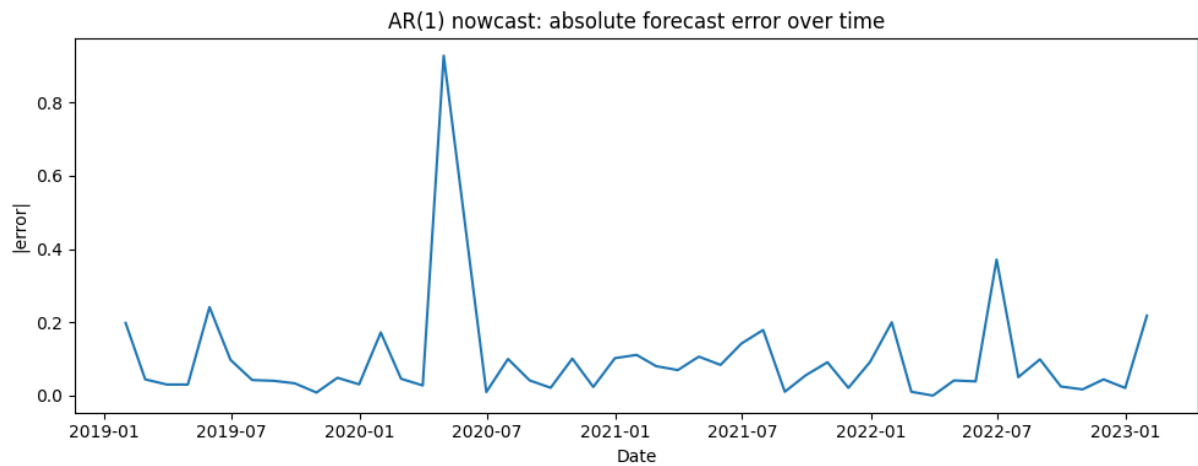


Figure 7: Residuals for cointegration

10.7 ARX nowcast design

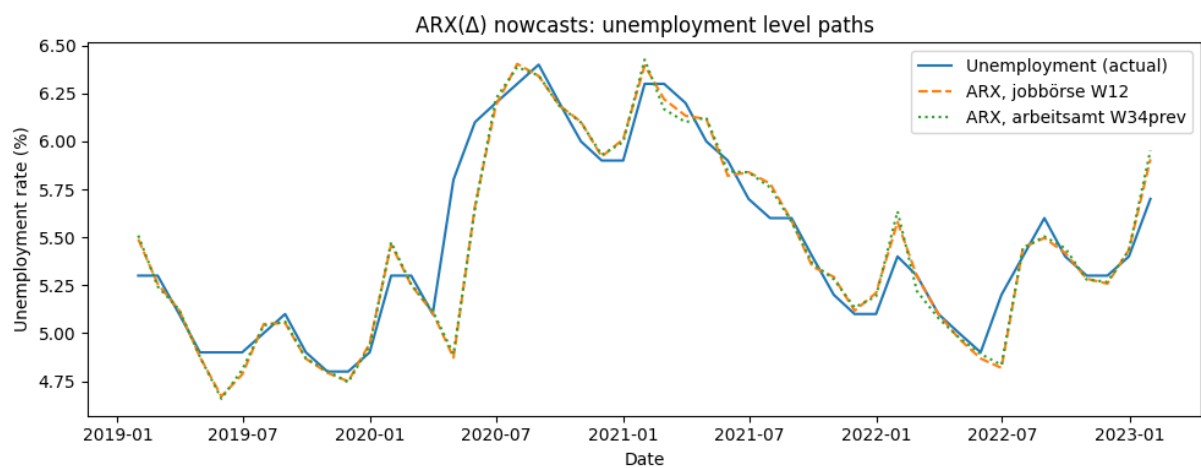


Figure 8: ARX nowcasts: unemployment level paths for jobbörse W12 and arbetsamt W34prev.

10.8 ARX absolute error comparison

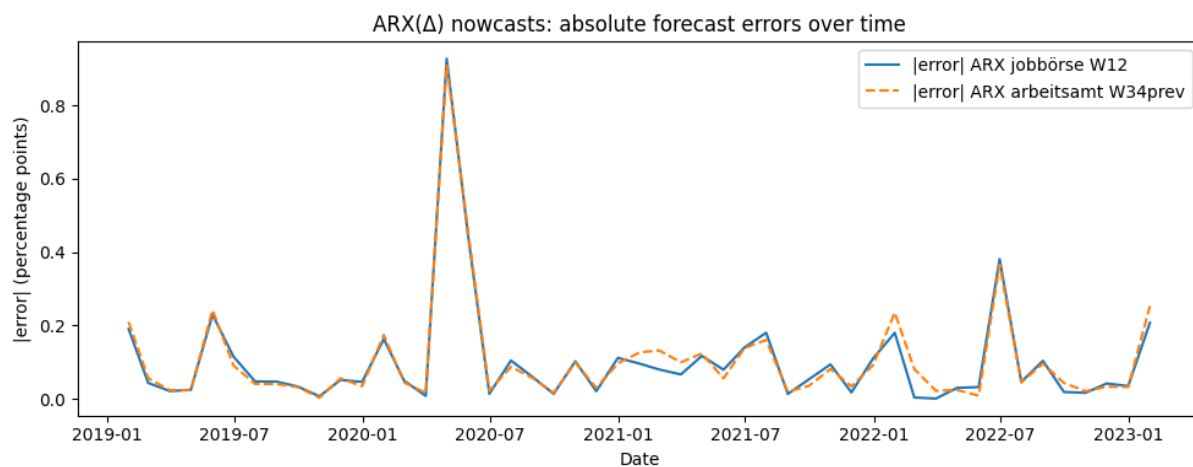


Figure 9: ARX nowcasts: absolute forecast errors over the evaluation sample.

10.9 ARX changes

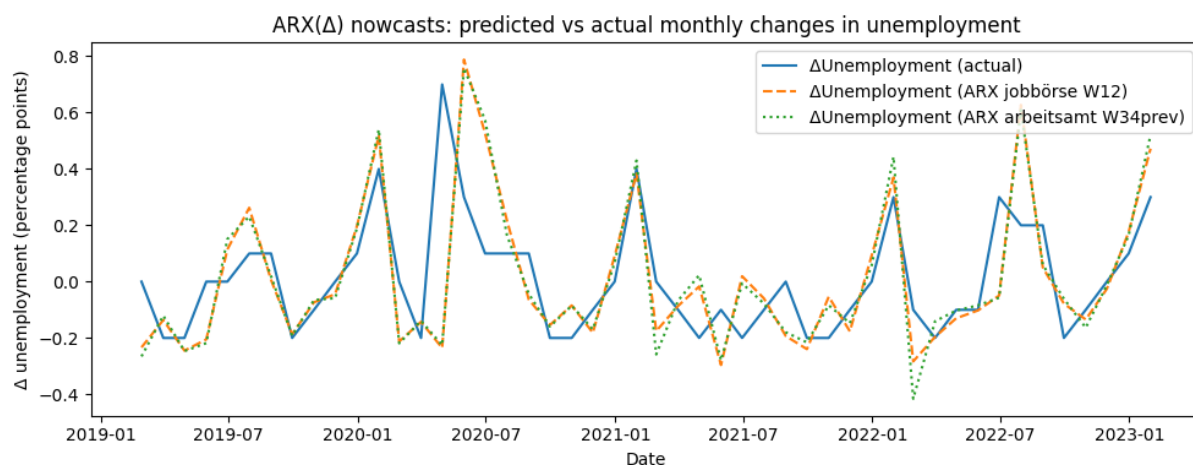


Figure 10: ARX nowcasts: predicted vs actual monthly changes in unemployment.

10.10 Midas level path

10.11 MIDAS absolute error

10.12 RMSE by month

10.13 Predicted vs actual

10.14 All models compared

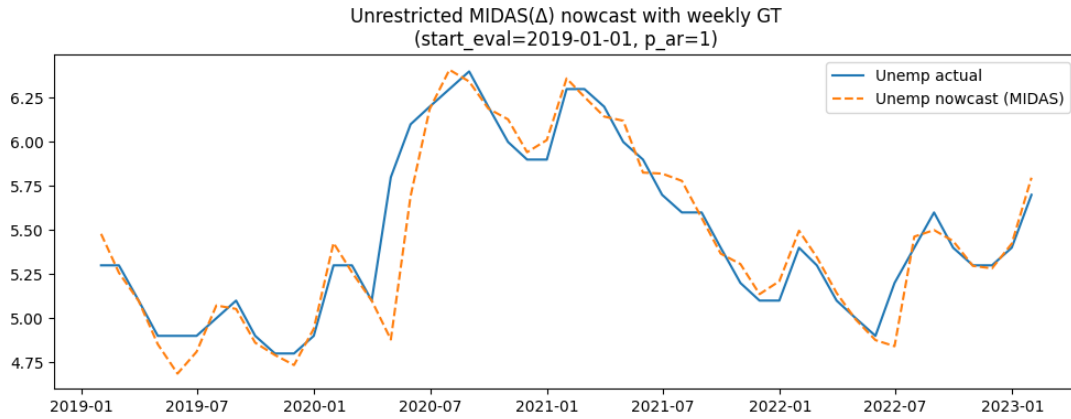


Figure 11: Unrestricted MIDAS nowcast with weekly jobbörse searches. The figure plots unemployment in levels (solid blue) and the U-MIDAS forecast (dashed orange) that combines eight weekly Google Trends lags with an autoregressive term in ΔU_t .

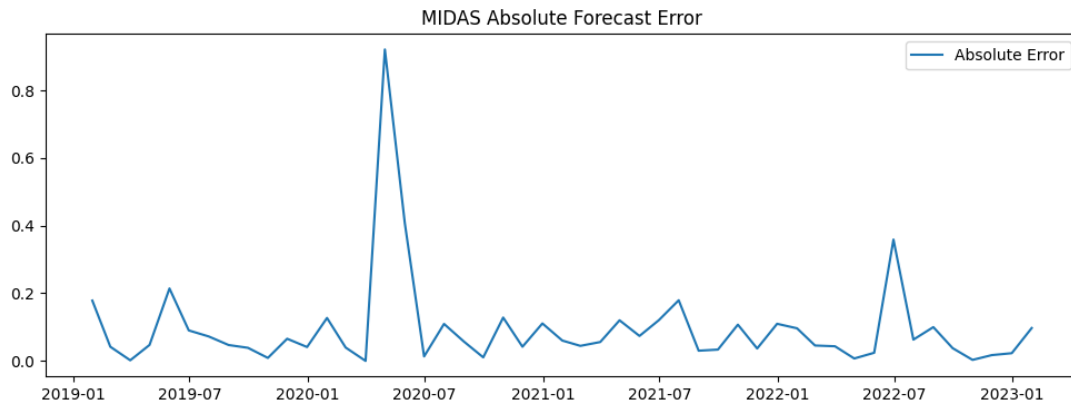


Figure 12: Absolute nowcast errors of the best U-MIDAS specification with weekly jobbörse searches. Errors are modest in normal times but spike sharply in spring 2020 and increase again in mid-2022.

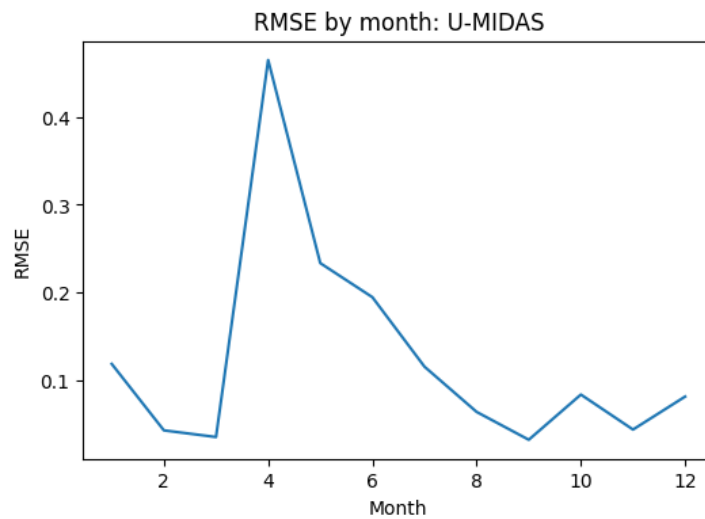


Figure 13: Root-mean-squared forecast error by calendar month for the best U-MIDAS model. The seasonal pattern closely mirrors that of the ARX specifications, with particularly large errors in April and May.

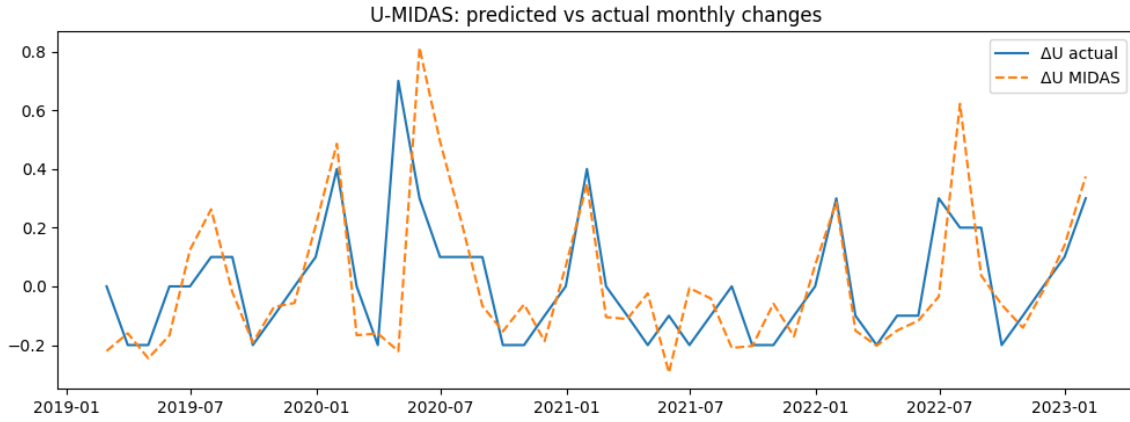


Figure 14: Predicted vs actual monthly changes in unemployment under the best U-MIDAS model. The figure shows realised ΔU_t and U-MIDAS predictions $\Delta \hat{U}_t$, indicating that weekly search data do not fully eliminate lagged responses around turning points.

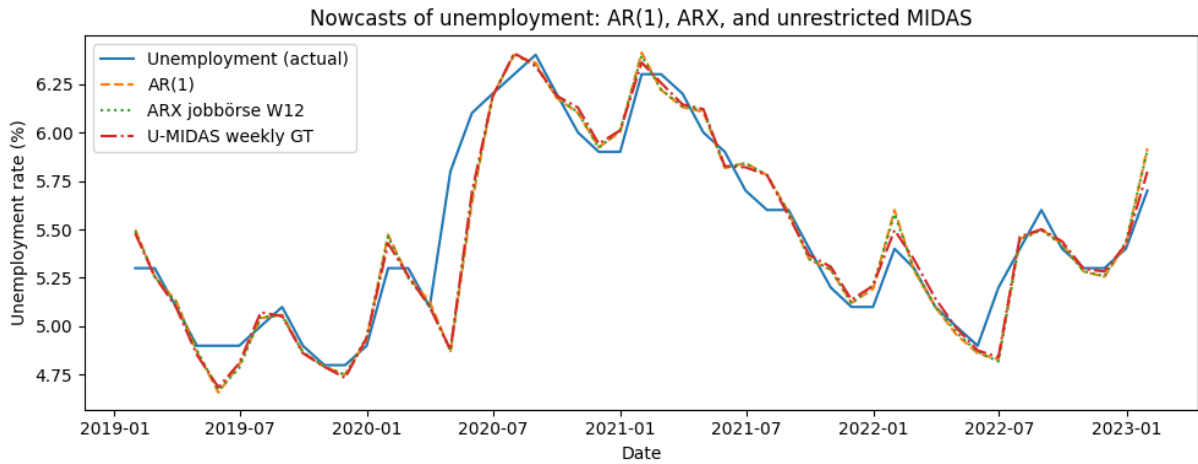


Figure 15: Real-time nowcasts of German unemployment from January 2019 to January 2023. The figure compares the benchmark AR(1) model in first differences with the best-performing ARX specification (jobbörse, weeks 1–2) and the best unrestricted MIDAS model with weekly jobbörse searches.

11 Appendix

11.1 Frequency of Google Trends

Table 9: Frequency of Google Trends data by requested time span

Requested time span	Example selection	Frequency returned
Up to ≈ 9 months	01.01.2022–30.09.2022	Daily index (one value per day)
Between ≈ 9 months and ≈ 5 years	01.01.2018–31.12.2022	Weekly index (one value per week)
Longer than ≈ 5 years	01.01.2004–31.12.2023	Monthly index (one value per month)

For a given keyword and region, the resolution of the Google Trends series depends on the requested time span. For short windows (up to roughly nine months) the public interface returns daily indices; for intermediate windows (up to roughly five years) weekly indices; and for longer windows monthly indices. This implies that requesting more than about five years of data at once will aggregate the underlying weekly information into monthly values, which is undesirable when constructing high-frequency predictors.

11.2 Stitching example

Table 10: Illustration of the stitching procedure for two overlapping windows

Week t	$x_t^{(j)}$	$x_t^{(j+1)}$	Ratio $x_t^{(j)} / x_t^{(j+1)}$	Rescaled $x_t^{(j+1)} \cdot s_{j+1 \leftarrow j}$
t_1	40	50	0.80	42.2
t_2	60	75	0.80	63.3
t_3	80	90	0.89	84.4
t_4	100	110	0.91	103.8
Median	$s_{j+1 \leftarrow j} = \text{median}\{0.80, 0.80, 0.89, 0.91\} = 0.845$			

11.3 Calendar example for the biweekly assignment

This appendix provides a concrete example of how weekly (Sunday) observations are assigned to the biweekly aggregates $W12_M$ and $W34_{M-1}$ using the “whole-week” rule described in the main text. The example avoids the three-Sundays edge case.

Consider $M = \text{June } 2022$. The relevant Sundays around this month are:

$\{2022-05-15, 2022-05-22, 2022-05-29, 2022-06-05, 2022-06-12, 2022-06-19, 2022-06-26\}$.

The assignment follows the definitions in Eqs. (??)–(??):

$$\mathcal{W}_{\text{Jun22}}^{12} = \{ 2022-06-05, 2022-06-12 \}, \quad \mathcal{W}_{\text{Jun22}}^{34,\text{prev}} = \{ 2022-05-22, 2022-05-29 \}.$$

Here, Sundays in the first half of June (days ≤ 15) enter $\text{W12}_{\text{Jun22}}$. Sundays after the 15th of May (the second half of $M-1$) enter $\text{W34}_{\text{May22}}$, which is aligned forward to June for prediction.

For a stitched weekly index $\{\tilde{x}_t\}$, the biweekly aggregates used in Eq. (??) evaluate to simple two-point averages:

$$\begin{aligned} X_{\text{Jun22}}^{12} &= \frac{1}{2} \left(\tilde{x}_{2022-06-05} + \tilde{x}_{2022-06-12} \right), \\ X_{\text{Jun22}}^{34,\text{prev}} &= \frac{1}{2} \left(\tilde{x}_{2022-05-22} + \tilde{x}_{2022-05-29} \right). \end{aligned}$$

For completeness, the Sundays 2022-06-19 and 2022-06-26 belong to the *second* half of June and therefore contribute to $\text{W34}_{\text{Jun22}}$, which is aligned to July ($M+1$) in the main biweekly panel.

Appendix A.1: ADF test cases

Following Hamilton (1994, Table 17.3), the ADF regression can be written in general form as

$$\Delta y_t = \alpha + \delta t + \rho y_{t-1} + \sum_{j=1}^{p-1} \zeta_j \Delta y_{t-j} + \varepsilon_t, \quad (11)$$

where α is an intercept, δt a deterministic linear trend and the lagged differences control for serial correlation. The unit-root null hypothesis is $H_0: \rho = 0$ against $H_1: \rho < 0$, with different deterministic cases obtained by restrictions on α and δ . For convenience, the main cases used in this study are summarised below.

Case 1 (no constant, no trend). ADF regression:

$$\Delta y_t = \rho y_{t-1} + \sum_{j=1}^{p-1} \zeta_j \Delta y_{t-j} + \varepsilon_t. \quad (12)$$

The hypotheses can be expressed either in terms of ρ or in terms of the implied AR(1) parameter $\phi = 1 + \rho$:

$$H_0: \rho = 0 \quad (\phi = 1, \text{unit root}) \quad \text{vs} \quad H_1: \rho < 0 \quad (|\phi| < 1, \text{stationary}).$$

Under H_0 , y_t follows a random walk; under H_1 , y_t is a stationary AR(1) (up to the augmenting lags in differences).

Case 2 (constant, no trend). ADF regression:

$$\Delta y_t = \alpha + \rho y_{t-1} + \sum_{j=1}^{p-1} \zeta_j \Delta y_{t-j} + \varepsilon_t. \quad (13)$$

The hypotheses are again $H_0: \rho = 0$ versus $H_1: \rho < 0$. In levels, this corresponds to

$$H_0: y_t = \mu + y_{t-1} + u_t \quad (\text{random walk with drift})$$

versus

$$H_1: y_t = \mu + \phi y_{t-1} + u_t, \quad |\phi| < 1 \quad (\text{stationary AR}(1) \text{ around a nonzero mean}).$$

Case 4 (constant and linear trend). ADF regression:

$$\Delta y_t = \alpha + \delta t + \rho y_{t-1} + \sum_{j=1}^{p-1} \zeta_j \Delta y_{t-j} + \varepsilon_t. \quad (14)$$

The null and alternative are again $H_0: \rho = 0$ versus $H_1: \rho < 0$. In levels, this corresponds to

$$H_0: y_t = \mu + \beta t + y_{t-1} + u_t \quad (\text{unit root with deterministic trend})$$

versus

$$H_1: y_t = \mu + \beta t + \phi y_{t-1} + u_t, \quad |\phi| < 1 \quad (\text{trend-stationary process}).$$

In the empirical application, the appropriate case is chosen based on visual inspection of the series (trend vs no trend), and the ADF test is implemented using the corresponding specification.

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